

Predictive Probability of Success (PoS)

Virtual ICB case study: Using PFS to predict OS ($\log(\text{HR})$, $p < 0.05$)

Internal Methodology Share

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Background: why PoS (not only power)

In drug development, a Go/No-Go decision often depends on the chance that a *future* confirmatory analysis will meet a pre-specified success criterion.

Classical power assumes the true treatment effect is fixed and known (or set to a design alternative). In practice, the true effect is uncertain and should be updated by accumulating evidence (historical trials, interim data, surrogate endpoints).

Predictive Probability of Success (PoS) answers:

“Given what we know today, what is the probability the future study will be declared successful?”

Core objects and notation

Let Y^* denote the test statistic or effect estimate from the *future* (or not-yet-mature) analysis, and let R be the **success region** defined by the protocol.

Examples of R : - Frequentist: $p < 0.05$ (two-sided) and effect in the favorable direction

- Bayesian: $\Pr(\theta < 0 \mid \text{data}) > 0.95$

- Clinical: lower CI bound exceeds a clinically meaningful threshold

Let θ denote the **true treatment effect** (e.g., log hazard ratio).

PoS definition: integral over uncertainty in the true effect

The building block is the conditional success probability:

$$\Pr(Y^* \in R \mid \theta).$$

Because θ is unknown, PoS averages the conditional success probability over the current posterior:

$$\text{PoS} = \int \Pr(Y^* \in R \mid \theta) p(\theta \mid Y_o) d\theta,$$

where Y_o denotes all evidence observed up to now (e.g., historical trials, interim information).

Equivalent view: posterior predictive distribution

An equivalent expression uses the posterior predictive distribution of the future result:

$$p(y^* | Y_o) = \int p(y^* | \theta) p(\theta | Y_o) d\theta.$$

Then PoS is simply:

$$\text{PoS} = \Pr(Y^* \in R | Y_o) = \int \mathbf{1}(y^* \in R) p(y^* | Y_o) dy^*.$$

This view is often convenient for computation and interpretation.

Success rule for $\log(\text{HR})$ with two-sided $p < 0.05$

Assume the future OS analysis estimates the log hazard ratio:

$$Y^* \mid \theta \sim N(\theta, \sigma_*^2),$$

and we target benefit with $HR < 1$ (i.e., $\theta < 0$).

For a two-sided test at $\alpha = 0.05$, the success region can be written as:

$$R = \{ Y^* < c \}, \quad c = -z_{1-\alpha/2} \sigma_* = -1.96 \sigma_*.$$

Therefore:

$$\text{PoS} = \Pr(Y^* < c \mid Y_o).$$

Linking the ICB surrogate scenario to PoS

In the ICB setting, PFS typically matures earlier than OS.

We use historical trials reporting both endpoints to learn the joint relationship:

$$(\theta_{OS}, \theta_{PFS})^\top \sim \text{MVN}(\mu, \Sigma_{\text{between}}),$$

and then update OS using the observed current-trial PFS result y_c .

This yields a predictive distribution for the final OS estimate:

$$Y_{OS}^* \mid y_c \sim N(m_{\text{post}}, v_{\text{post}} + \sigma_*^2),$$

and PoS follows from the OS success rule:

$$\text{PoS} = \Pr(Y_{OS}^* < -1.96 \sigma_* \mid y_c).$$

What we're trying to answer

Scenario (ICB trial):

- ▶ Primary endpoint: **PFS** (read out earlier)
- ▶ Key secondary endpoint: **OS** (matures later)
- ▶ Effect measure: **log(HR)** for both endpoints

Decision question:

Given the current trial's observed PFS result, what is the probability that the final OS analysis will be statistically significant (two-sided $p < 0.05$) in the beneficial direction?

PoS in one line

Let:

- ▶ θ = the (unknown) *true* treatment effect (here: $\theta = \log(HR)$)
- ▶ Y^* = the future study's estimator (e.g., estimated $\log(HR)$ at final OS analysis)
- ▶ R = the **success region** (defined by your decision rule)

Then predictive probability of success is:

$$\text{PoS} = \int \Pr(Y^* \in R \mid \theta) \pi(\theta \mid \text{current evidence}) d\theta$$

Interpretation: average the conditional “power” over uncertainty in the true effect.

Our success rule ($\log(HR)$, two-sided $p < 0.05$)

For OS final analysis, assume:

$$Y_{OS}^* \mid \theta_{OS}^* \sim \mathcal{N}(\theta_{OS}^*, se_*^2)$$

Two-sided $\alpha = 0.05$, beneficial direction $HR < 1$ (so $\log(HR) < 0$):

$$R = \{Y_{OS}^* < c\}, \quad c = -z_{0.975} se_* \approx -1.96 se_*$$

So:

$$\text{PoS} = \Pr(Y_{OS}^* < c \mid \text{current evidence})$$

Where “current evidence” comes from

We will combine two sources:

1) **Historical trials** (e.g., 27 ICB studies) that report both endpoints

▶ $y_{i,OS} = \log(HR_{OS,i}), se_{i,OS}$

▶ $y_{i,PFS} = \log(HR_{PFS,i}), se_{i,PFS}$

2) **Current trial's PFS** (available now)

▶ $y_c = \log(HR_{PFS}), se_c$

Goal: obtain $\theta_{OS}^* \mid y_c$ (predictive distribution for the current trial's *true* OS effect), then compute PoS.

Model (concept picture)

Two layers:

- ▶ **Between-study layer (true effects differ across trials):**

$$(\theta_{i,OS}, \theta_{i,PFS})^\top \sim \text{MVN}(\mu, \Sigma_{between})$$

- ▶ **Within-study layer (estimation error):**

$$(y_{i,OS}, y_{i,PFS})^\top \sim \text{MVN}((\theta_{i,OS}, \theta_{i,PFS})^\top, \Sigma_{within,i})$$

Key parameters:

- ▶ μ_{OS}, μ_{PFS} : average (logHR) effects across trials
- ▶ τ_{OS}, τ_{PFS} : between-study SDs (heterogeneity)
- ▶ ρ : correlation between true OS and true PFS effects

Conditioning: OS effect given observed PFS

Treat the current trial's true effects $(\theta_{OS}^*, \theta_{PFS}^*)$ as a draw from the same between-study distribution.

Observation model for current PFS:

$$y_c = \theta_{PFS}^* + e, \quad e \sim \mathcal{N}(0, se_c^2)$$

If $(\theta_{OS}^*, \theta_{PFS}^*) \sim \text{MVN}(\mu, \Sigma_{\text{between}})$, then:

$$\theta_{OS}^* \mid y_c \sim \mathcal{N}(m_{\text{post}}, v_{\text{post}})$$

where

$$m_{\text{post}} = \mu_{OS} + \frac{\text{Cov}(OS, PFS)}{\text{Var}(PFS) + se_c^2} (y_c - \mu_{PFS})$$

$$v_{\text{post}} = \text{Var}(OS) - \frac{\text{Cov}(OS, PFS)^2}{\text{Var}(PFS) + se_c^2}$$

PoS in closed form

Final OS estimator:

$$Y_{OS}^* | y_c \sim \mathcal{N}(m_{post}, v_{post} + se_*^2)$$

Therefore:

$$\text{PoS} = \Pr(Y_{OS}^* < c | y_c) = \Phi\left(\frac{c - m_{post}}{\sqrt{v_{post} + se_*^2}}\right)$$

Knobs that matter:

- ▶ ρ : stronger PFS–OS association $\rightarrow$ PFS updates OS more
- ▶ τ 's: more heterogeneity $\rightarrow$ less certainty $\rightarrow$ PoS often down
- ▶ d_{events} : more OS events $\rightarrow$ smaller se_* $\rightarrow$ PoS up

Reproducible demo: simulate 27 historical ICB trials

We'll create a virtual dataset with:

- ▶ $K = 27$ historical trials
- ▶ True bivariate relationship between $\log(\text{HR_OS})$ and $\log(\text{HR_PFS})$
- ▶ Realistic ranges of standard errors

Estimate the between-study relationship (demo-friendly moment approach)

For a quick internal demo, we'll use **error-corrected moments**:

- ▶ observed sample covariance: $S_{obs} = \text{Cov}(y)$
- ▶ average within-study covariance: $\bar{W}$
- ▶ estimate: $\hat{\Sigma}_{between} \approx S_{obs} - \bar{W}$

```
$mu_hat
```

	OS	PFS
	-0.1725548	-0.2952081

```
$Sigma_between_hat
```

	OS	PFS
OS	0.01509471	0.01329543
PFS	0.01329543	0.01171239

```
$tau_hat
```

	OS	PFS
	0.1228605	0.1082238

```
$rho_hat
```


Current trial: plug in observed PFS and compute PoS

Assume the current trial has interim/final PFS result:

- ▶ $y_c = \log(HR_{PFS})$
- ▶ se_c (its standard error)

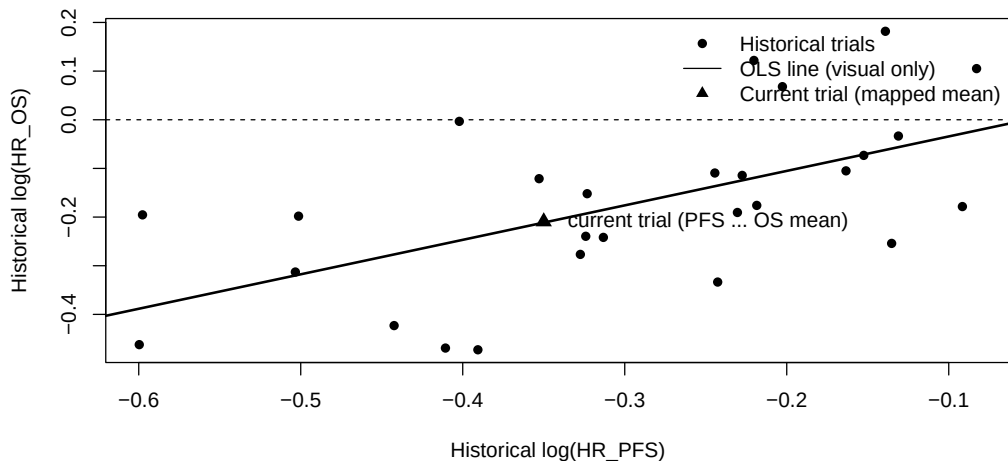
For OS final analysis, assume planned OS event count d . Under 1:1 allocation, Schoenfeld gives a useful approximation:

$$se_* \approx \frac{2}{\sqrt{d}}$$

<code>y_c</code>	<code>HR_PFS</code>	<code>m_post_OS.OS</code>
-0.35000000	0.70468809	-0.20932372
<code>HR_OS_pred_mean.OS</code>	<code>sd_thetaOS</code>	<code>d_events</code>
0.81113261	0.07856589	240.00000000
<code>se_star</code>	<code>threshold_c</code>	<code>PoS</code>
0.12909944	-0.25303026	0.38621237

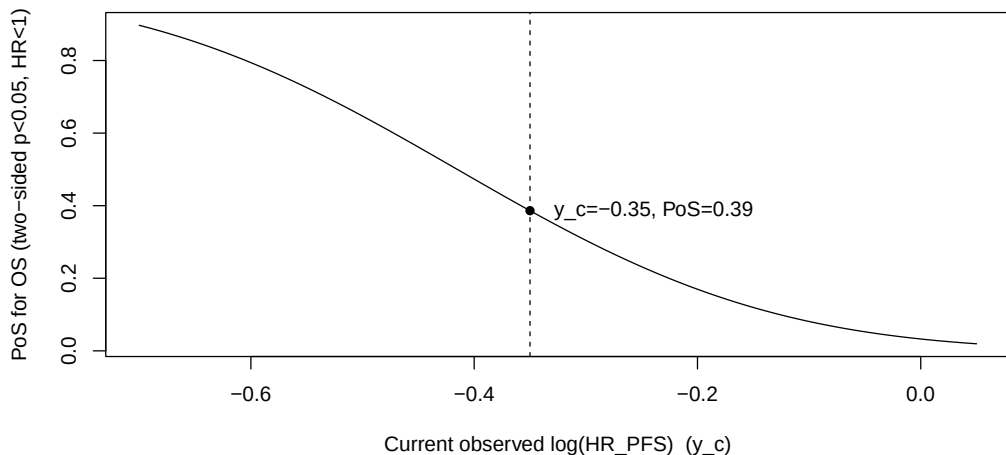
Visual 1: historical relationship (virtual 27 trials)

Virtual 27 ICB trials: $\log(\text{HR_PFS})$ vs $\log(\text{HR_OS})$



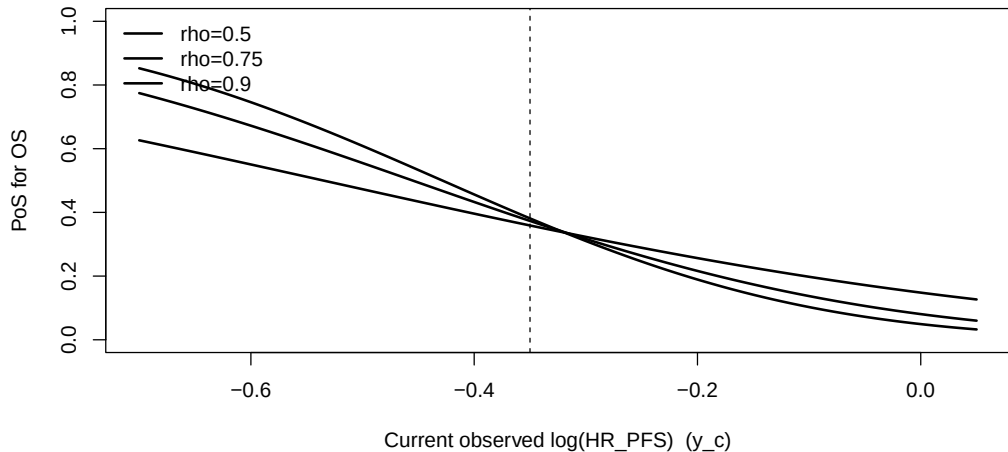
Visual 2: PoS curve (how PFS maps to OS success probability)

PoS curve: PFS result ... OS success probability



Sensitivity demo: vary PFS–OS correlation (ρ)

Sensitivity: PoS curve under different rho



Sensitivity demo: OS information (event count) matters

If your event count changes, so does se_* and the success threshold c .

	d_events	se_star	threshold_c	PoS
1	150	0.1632993	-0.3200608	0.2705742
2	200	0.1414214	-0.2771808	0.3374467
3	240	0.1290994	-0.2530303	0.3862124
4	300	0.1154701	-0.2263171	0.4515786
5	400	0.1000000	-0.1959964	0.5417320

Takeaways (what to emphasize internally)

- ▶ PoS answers a **decision** question: *probability of meeting a success rule*, not “best guess HR”.
- ▶ In this ICB setup:
 - ▶ Historical trials estimate a **between-study** OS–PFS association (ρ , heterogeneity)
 - ▶ Current PFS readout updates the predicted **true OS effect** via conditional normal formulas
 - ▶ Final OS success probability is a **one-line Normal CDF**
- ▶ Sensitivity analysis is not optional:
 - ▶ ρ (association strength), heterogeneity (τ), and OS information (event count) are the big levers.

Next steps (toward production)

If this moves beyond a virtual demo:

- ▶ Fit a full bivariate random-effects model (REML/Bayesian)
- ▶ Treat within-study correlation r as unknown $\rightarrow$ range or prior
- ▶ Add design features that affect OS–PFS link (crossover, line of therapy, combo vs mono)
- ▶ Use the same framework for interim decisions: re-run PoS as PFS/OS mature

How to run this deck

- 1) Save this file as `PoS_virtual_ICB_demo.qmd`
- 2) In RStudio / VS Code terminal:

```
quarto render PoS_virtual_ICB_demo.qmd
```

It will produce an HTML slide deck (RevealJS) with all plots and numbers.